

Kiran Mohan

Assignment 1 :

Internship Assignment

Cyber Security and Digital Forensics

Assignment 1:Reconnaissance

- <https://tryhackme.com/room/passiverecon>

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tryhackme.com
TryHackMe | Passive Reconnaissance

It requires. This room requires basic networking knowledge along with basic familiarity with the command line. The module provide the required knowledge if necessary.

Woop woop! Your answer is correct

Important Notice: Please note that if you're not subscribed, the AttackBox won't have Internet access, so you will need to use the VPN to complete the questions that require Internet access.

Answer the questions below

This room does not use a target virtual machine (VM) to demonstrate the discussed topics. Instead, we will query public WHOIS servers and DNS servers for domains owned by TryHackMe. Start the AttackBox and make sure it is ready. You will use the AttackBox to answer the questions in later tasks, especially tasks 3 and 4.

No answer needed

Correct Answer

Task 2 ☐ Passive Versus Active Recon

Task 3 ☐ Whois

Task 4 ☐ nslookup and dig

Task 5 ☐ DNSDumpster

Task 6 ☐ Shodan.io

Task 7 ☐ Summary

Created by
tryhackme strategos Aashir.Masood

Room Type
Free Room. Anyone can deploy virtual machines

Users in Room
180,759

Created
1280 days ago

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TryHackMe | Passive Reconnaissance

• Entering company premises pretending to be a repairman.

Considering the invasive nature of active reconnaissance, one can quickly get into legal trouble unless one obtains proper legal

Woop woop! Your answer is correct

Answer the questions below

You visit the Facebook page of the target company, hoping to get some of their employee names. What kind of reconnaissance activity is this? (A for active, P for passive)

P

Correct Answer

You ping the IP address of the company webserver to check if ICMP traffic is blocked. What kind of reconnaissance activity is this? (A for active, P for passive)

A

Correct Answer

You happen to meet the IT administrator of the target company at a party. You try to use social engineering to get more information about their systems and network infrastructure. What kind of reconnaissance activity is this? (A for active, P for passive)

A

Correct Answer

Task 3 ☐ Whois

Task 4 ☐ nslookup and dig

Task 5 ☐ DNSDumpster

Task 6 ☐ Shodan.io

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Information collected can be inspected to find new attack surfaces, such as social engineering or technical attacks. For instance, an attacker might consider an attack against the email server of the admin user or the DNS servers, assuming they are owned by your client.

It is important to note that due to automated tools abusing WHOIS queries to harvest email addresses, many WHOIS services take measures against this. They might redact email addresses, for instance. Moreover, many registrants subscribe to privacy services to avoid their email addresses being harvested by spammers and keep their information private.

On the AttackBox, open the terminal and run the `whois tryhackme.com` command to get the information you need to answer the following questions.

Answer the questions below

When was TryHackMe.com registered?

20180705 ✓ Correct Answer 🔍 Hint

What is the registrar of TryHackMe.com?

namecheap.com ✓ Correct Answer 🔍 Hint

Which company is TryHackMe.com using for name servers?

cloudflare.com ✓ Correct Answer 🔍 Hint

Task 4 ☐ nslookup and dig

Task 5 ☐ DNSDumpster

Task 6 ☒ Shodan.io

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• SERVER is the DNS server that you want to query.
• DOMAIN_NAME is the domain name you are looking up.
• TYPE contains the DNS record type, as shown in the table provided earlier.

✓ Woop woop! Your answer is correct

```
user@TryHackMe$ dig tryhackme.com MX
; <>> DiG 9.16.19-RH <>> tryhackme.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<
```

A quick comparison between the output of `nslookup` and `dig` shows that `dig` returned more information, such as the TTL (Time To Live) by default. If you want to query a `1.1.1.1` DNS server, you can execute `dig @1.1.1.1 tryhackme.com MX`.

Using the AttackBox, open the terminal and use the `nslookup` or `dig` command to get the information you need to answer the following question.

Answer the questions below

Check the TXT records of thmlabs.com. What is the flag there?

THM{a5b83929888ed36acb0272971e438d78} ✓ Correct Answer

Task 5 ☐ DNSDumpster

Hey, Congratulations! You submitted the correct answer for task 4 question 1. Keep it up!

Safari File Edit View History Bookmarks Window Help tryhackme.com

TryHackMe | Passive Reconnaissance DNSDumpster - Find & lookup dns records for recon & research

173.194.206.26 64.233.186.104 209.85.202.26

1 aspmx.l.google.com. 5 alt1.aspmx.l.google.com. 5 alt2.aspmx.l.google.com.

✓ Woop woop! Your answer is correct

Use the web browser on the AttackBox, or your system, to answer the following question.

Answer the questions below

Lookup tryhackme.com on DNSDumpster. What is one interesting subdomain that you would discover in addition to www and blog?

remote ✓ Correct Answer

Task 6 ○ Shodan.io

Task 7 ○ Summary

How likely are you to recommend this room to others?

Hey, Congratulations! You submitted the correct answer for task 4 question 1. Keep it up!

2 3 4 5 6 7 8 9 10

Submit now

Safari File Edit View History Bookmarks Window Help tryhackme.com

TryHackMe | Passive Reconnaissance DNSDumpster - Find & lookup dns records for recon & research nginx - Shodan Search

You may also try searching for the IP addresses you have obtained from DNS lookups. These are, of course, more subject to change than the domain names, and you are encouraged to join TryHackMe's [Shodan.io](#).

It would be best to visit Shodan.io to answer the following questions; however, note that you can find the answers on Shodan.io without needing a premium account.

Answer the questions below

According to Shodan.io, what is the 2nd country in the world in terms of the number of publicly accessible Apache servers?

China ✓ Correct Answer Hint

Based on Shodan.io, what is the 3rd most common port used for Apache?

8080 ✓ Correct Answer Hint

Based on Shodan.io, what is the 3rd most common port used for nginx?

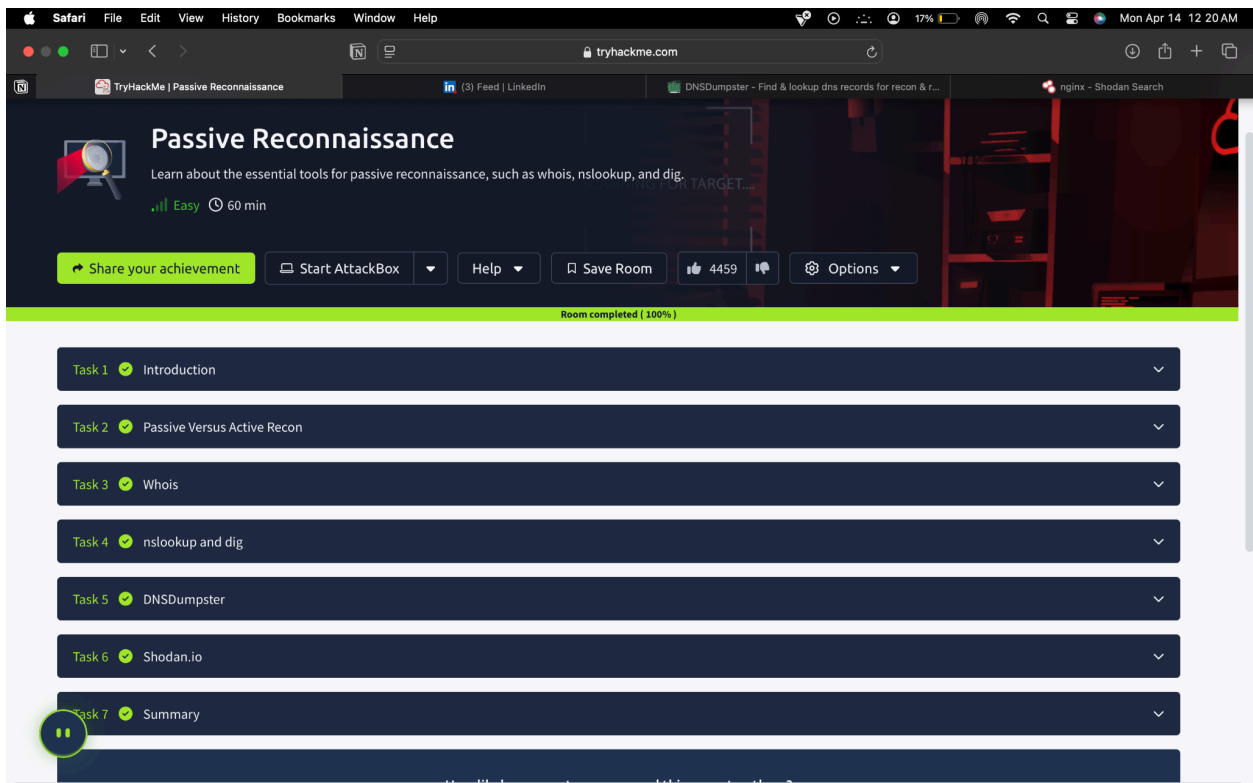
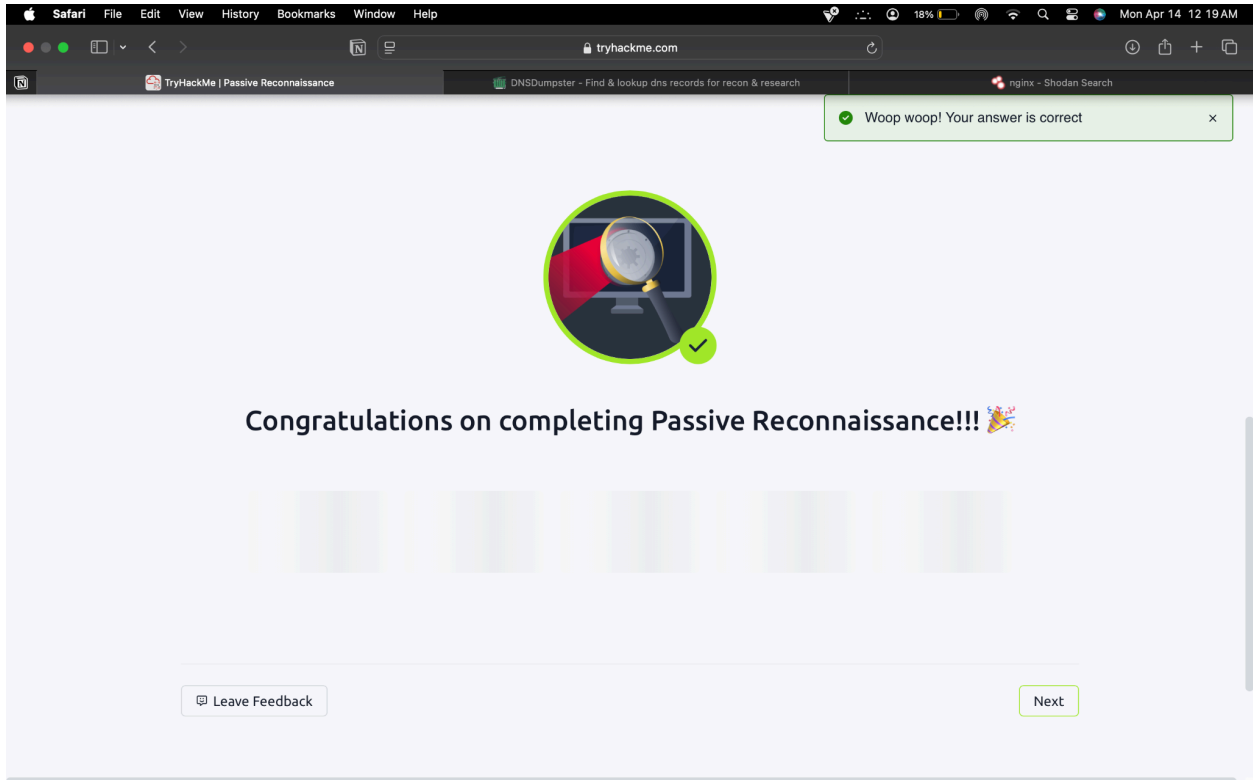
5000 ✓ Correct Answer Hint

Task 7 ○ Summary

How likely are you to recommend this room to others?

1 2 3 4 5 6 7 8 9 10

Submit now



- <https://tryhackme.com/room/activerecon>

The screenshot displays the TryHackMe web interface in a Safari browser. The browser's address bar shows the URL `tryhackme.com`. The page title is "TryHackMe | Active Reconnaissance".

The main content area on the left contains the following text:

can be "armed" to become an efficient reconnaissance framework. Afterwards, we discuss other benign tools such as `ping`, `traceroute`, and `telnet`. All these programs require connection to the target, and hence our activities would fall under active reconnaissance.

This room is of interest to anyone who wants to become familiar with essential tools and see how they can use them in active reconnaissance. The web browser developer tools might take some effort to gain familiarity, although it offers a graphical user interface. The command-line tools covered are relatively straightforward to use.

Important Notice: Please note that if you're not subscribed, the AttackBox won't have Internet access, so you will need to use the [VPN](#) to complete the questions that require Internet access.

Answer the questions below

Ensure that you understand why these tools fall under active reconnaissance. Launch your AttackBox and ensure that it is ready. You will need it to answer the questions, especially in later tasks.

Below the text are two buttons: "No answer needed" and "✓ Correct Answer".

A task list is visible on the left side of the interface:

- Task 2 ○ Web Browser
- Task 3 ○ Ping
- Task 4 ○ Traceroute
- Task 5 ● Telnet (highlighted with a green circle)
- Task 6 ○ Netcat

The right pane shows a dark-themed virtual machine interface with the text "Your machine is initializing..." and "Use the AttackBox to attack machines you start on tasks". A progress bar indicates "Loading (9%)". At the bottom of the right pane, it says "Access desktop in 109s" and "THM AttackBox 59min 48s".

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tryhackme.com

TryHackMe | Active Reconnaissance

Woop woop! Your answer is correct

static-labs.tryhackme.cloud/sites/

TryHackMe | Learn Cy... TryHackMe Support Offline CyberChef

Over time, you might find a few extensions that fit perfectly in your workflow.

Answer the questions below

Browse to the following website and ensure that you have opened your Developer Tools on AttackBox Firefox, or the browser on your computer. Using the Developer Tools, figure out the total number of questions.

8

Correct Answer Hint

Task 3 Ping

Task 4 Traceroute

Task 5 Telnet

Task 6 Netcat

Task 7 Putting It All Together

THM AttackBox 45min 25s

TCP/IP Chat

https://static-labs.tryhackme.cloud/sites/networking-tcp/

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec

ACK : Erm... What?

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Sources Outline Search

script.js

```
49 'answer_3' : 'FIN/ACK : Yeah Me Too',
50 'answer' : 3
51 },
52 8 : {
53   'speaking' : 'alice',
54   'answer_1' : 'SYN : Connected',
55   'answer_2' : 'ACK : Okay, Goodbye',
56   'answer_3' : 'SYN/ACK : Not Received',
57   'answer' : 2
58 }
59
```

Watch expressions

Breakpoints

Pause on debugger statement

Pause on exceptions

XHR Breakpoints

Event Listener Breakpoints

DOM Mutation Breakpoints

Log

(59, 2)

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tryhackme.com

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nginx - Shodan Search

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ping by default.

- Your system is unplugged from the network.

Answer the questions below

Which option would you use to set the size of the data carried by the ICMP echo request?

-s ✓ Correct Answer Hint

What is the size of the ICMP header in bytes?

8 ✓ Correct Answer Hint

Does MS Windows Firewall block ping by default? (Y/N)

Y ✓ Correct Answer

Deploy the VM for this task and using the AttackBox terminal, issue the command `ping -c 10 10.10.118.68`. How many ping replies did you get back?

10 ✓ Correct Answer

Task 4 Traceroute

Task 5 Telnet

Task 6 Netcat

root's Home

root@ip-10-10-206-139: ~

```
File Edit View Search Terminal Help
18 bytes from 10.10.118.68: icmp_seq=20 ttl=64
18 bytes from 10.10.118.68: icmp_seq=21 ttl=64
18 bytes from 10.10.118.68: icmp_seq=22 ttl=64
18 bytes from 10.10.118.68: icmp_seq=23 ttl=64
18 bytes from 10.10.118.68: icmp_seq=24 ttl=64
^C
--- 10.10.118.68 ping statistics ---
24 packets transmitted, 24 received, 0% packet loss, time
ms
root@ip-10-10-206-139:~# ping -s 100 10.10.118.68
PING 10.10.118.68 (10.10.118.68) 100(128) bytes of data:
100 bytes from 10.10.118.68: icmp_seq=1 ttl=64 time=0.580 ms
100 bytes from 10.10.118.68: icmp_seq=2 ttl=64 time=0.760 ms
100 bytes from 10.10.118.68: icmp_seq=3 ttl=64 time=0.470 ms
100 bytes from 10.10.118.68: icmp_seq=4 ttl=64 time=2.64 ms
100 bytes from 10.10.118.68: icmp_seq=5 ttl=64 time=0.361 ms
100 bytes from 10.10.118.68: icmp_seq=6 ttl=64 time=0.526 ms
100 bytes from 10.10.118.68: icmp_seq=7 ttl=64 time=0.653 ms
^C
--- 10.10.118.68 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 610ms
rtt min/avg/max/mdev = 0.361/0.843/2.637/0.741 ms
root@ip-10-10-206-139:~#
```

THM AttackBox 25min 50s

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tryhackme.com

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TryHackMe | Active Reconnaissance

Some routers return a public IP address. You might examine a few of these routers based on the scope of the intended penetration testing.

- Some routers don't return a reply.

Answer the questions below

In Traceroute A, what is the IP address of the last router/hop before reaching tryhackme.com?

172.67.69.208 ✓ Correct Answer Hint

In Traceroute B, what is the IP address of the last router/hop before reaching tryhackme.com?

104.26.11.229 ✓ Correct Answer Hint

In Traceroute B, how many routers are between the two systems?

26 ✓ Correct Answer

Start the attached VM from Task 3 if it is not already started. On the AttackBox, run `tracert 10.10.118.68`. Check how many routers/hops are there between the AttackBox and the target VM.

No answer needed ✓ Correct Answer Hint

Task 5 Telnet

Task 6 Netcat

root's Home

root@ip-10-10-206-139: ~

```
File Edit View Search Terminal Help
28 * * *
29 * * *
30 * * *
root@ip-10-10-206-139:~# cat /etc/os-release
NAME="Ubuntu"
VERSION="20.04.6 LTS (Focal Fossa)"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 20.04.6 LTS"
VERSION_ID="20.04"
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
VERSION_CODENAME=focal
UBUNTU_CODENAME=focal
root@ip-10-10-206-139:~#
root@ip-10-10-206-139:~#
root@ip-10-10-206-139:~# traceroute 10.10.118.68
traceroute to 10.10.118.68 (10.10.118.68), 30 hops max, 60 byte packets
 1 10.10.118.68 (10.10.118.68) 0.765 ms 0.704 ms 0.671 ms
root@ip-10-10-206-139:~#
```

THM AttackBox 13min 14s

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tryhackme.com

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Woop woop! Your answer is correct

root's Home

root@ip-10-10-206-139: ~

File Edit View Search Terminal Help

```
root@ip-10-10-206-139:~# traceroute 10.10.118.68
traceroute to 10.10.118.68 (10.10.118.68), 30 hops max, 60 byte packets
 1  10.10.118.68 (10.10.118.68)  0.765 ms  0.704 ms  0.671 ms
root@ip-10-10-206-139:~#
root@ip-10-10-206-139:~#
root@ip-10-10-206-139:~#
root@ip-10-10-206-139:~#
root@ip-10-10-206-139:~# telnet 10.10.118.68 80
Trying 10.10.118.68...
Connected to 10.10.118.68.
Escape character is '^['.
GET / HTTP/1.1
HTTP/1.1 408 Request Timeout
Date: Sun, 13 Apr 2025 20:11:59 GMT
Server: Apache/2.4.61 (Debian)
Content-Length: 329
Connection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>408 Request Timeout</title>
</head><body>
<h1>Request Timeout</h1>
<p>Server timeout waiting for the HTTP request from the client.</p>
<hr>
<address>Apache/2.4.61 (Debian) Server at ip-10-10-118-68.eu-west-1.compute.internal Port 80</address>
</body></html>
Connection closed by foreign host.
root@ip-10-10-206-139:~#
```

Of particular interest for us is discovering the type and version of the installed web server, **Server: nginx/1.6.2**. In this example, we communicated with a web server, so we used basic HTTP commands. If we connect to a mail server, we need to use proper commands based on the protocol, such as SMTP and POP3.

Answer the questions below

Start the attached VM from Task 3 if it is not already started. On the AttackBox, open the terminal and use the telnet client to connect to the VM on port 80. What is the name of the running server?

Apache ✓ Correct Answer

What is the version of the running server (on port 80 of the VM)?

2.4.61 ✓ Correct Answer

Task 6 Netcat

Task 7 Putting It All Together

Hey, Congratulations! You submitted the correct answer for task 5 question 2. Keep it up!

Room to others?

THM AttackBox 3min 26s

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tryhackme.com

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nginx - Shodan Search

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Woop woop! Your answer is correct

root's Home

root@ip-10-10-206-139: ~

File Edit View Search Terminal Help

```
root@ip-10-10-206-139:~# nc 10.10.118.68 80
GET / HTTP/1.1
HTTP/1.1 400 Bad Request
Date: Sun, 13 Apr 2025 20:15:55 GMT
Server: Apache/2.4.61 (Debian)
Content-Length: 334
Connection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>400 Bad Request</title>
</head><body>
<h1>Bad Request</h1>
<p>Your browser sent a request that this server could not understand.<br />
</p>
<hr>
<address>Apache/2.4.61 (Debian) Server at ip-10-10-118-68.eu-west-1.compute.internal Port 80</address>
</body></html>
```

has the IP address **10.10.118.68** so we can connect to it from the client-side by executing **nc 10.10.118.68 1234**. This setup would echo whatever you type on one side to the other side of the TCP tunnel. You can find a recording of the process below. Note that the listening server is on the left side of the screen.

Answer the questions below

Start the VM and open the AttackBox. Once the AttackBox loads, use Netcat to connect to the VM port 21. What is the version of the running server?

0.17 ✓ Correct Answer

Task 7 Putting It All Together

How likely are you to recommend this room to others?

1 2 3 4 5 6 7 8 9 10

Submit now

Created by

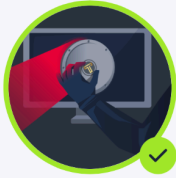
tryhackme strategos

Room Type Users in Room Created

THM AttackBox 0m 0s

tryhackme.com

Woop woop! Your answer is correct



Congratulations on completing Active Reconnaissance!!! 🎉

Points earned 88	Completed tasks 7	Room type Walkthrough	Difficulty Easy	Streak 1
---------------------	----------------------	--------------------------	--------------------	-------------

Leave Feedback

Next

Safari File Edit View History Bookmarks Window Help

tryhackme.com

Share your achievement Start AttackBox Help Save Room 3086 Options

Room completed (100%)

Target Machine Information

Title	Target IP Address	Expires	
NetSecMod Room 02 telnet-cibadr	10.10.118.68	7min 42s	? Add 1 hour Terminate

- Task 1 Introduction
- Task 2 Web Browser
- Task 3 Ping
- Task 4 Traceroute
- Task 5 Telnet
- Task 6 Netcat
- Task 7 Putting It All Together

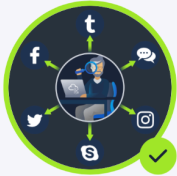
- <https://tryhackme.com/room/ohsint>

Safari File Edit View History Bookmarks Window Help

tryhackme.com

TryHackMe | OhSINT owoodflint - Google Search OWoodflint/people_finder... (3) 0x0000000000000000... owoodflint - Oliver Woodflint... WIGLE: Wireless Network... Start Page

✓ Woop woop! Your answer is correct



Congratulations on completing OhSINT!!! 🎉

Points earned 🔥 210	Completed tasks ✅ 1	Room type 🚩 Challenge	Difficulty 📶 Easy	Streak 🔥 1
------------------------	------------------------	--------------------------	----------------------	---------------

🗉 Leave Feedback

Next

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tryhackme.com

TryHackMe | OhSINT Feed | LinkedIn owoodflint - Google S... OWoodflint/people_fi... (3) 0x0000000000000000... owoodflint - Oliver W... WIGLE: Wireless Netw... Start Page

Answer the questions below

What is this user's avatar?

cat

✓ Correct Answer Hint

What city is this person in?

London

✓ Correct Answer Hint

What is the SSID of the WAP he connected to?

UnileverWiFi

✓ Correct Answer

What is his personal email address?

OWoodflint@gmail.com

✓ Correct Answer

What site did you find his email address on?

Github

✓ Correct Answer

Where has he gone on holiday?

New York

✓ Correct Answer Hint

What is the person's password?

pennYDröpper!

✓ Correct Answer Hint